Applied Thermal Engineering 213 (2022) 118663

Contents lists available at ScienceDirect

Applied Thermal Engineering

journal homepage: www.elsevier.com/locate/apthermeng

Research Paper

Extended state observer-based model predictive temperature control of mechanically pumped two-phase loop: An experimental study

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ARTICLE INFO

Keywords:
Two-phase loop
Extended state observer (ESO)
Model predictive control (MPC)
Temperature control

ABSTRACT

Mechanically pumped two-phase loop (MPTL), as an emerging two-phase heat dissipation transformational technique, has made booming progress in some critical but tough cases such as space stations and high-performance chips. However, accurate temperature regulation of MPTL is intractable due to periodic heat load disturbance, measurement noise, and micro-channel boiling nonlinearity. To address these issues, an extended state observer-based model predictive control (ESOMPC) strategy is developed in this paper. To describe the process model, this paper builds an MPTL platform equipped with a micro-channel evaporator, based on which a series of sinusoidal heat load variation is carried out to show the effects on the wall temperature. Open-loop step experiments are then conducted in terms of flow rate of the working fluid so that the process model can be identified and the nonlinearity can be analyzed. Then, the ESOMPC is developed by incorporating the disturbance information in the optimization framework of the MPC. The ESO part is utilized to estimate the unknown disturbances and compensate the nonlinearity of MPTL, and thus enhance the disturbance mitigation property of the MPC algorithm. Finally, the closed-loop experiments verifies the efficacy of the proposed solution, showing smaller temperature fluctuation and shorter settling time than the conventional controller. The results indicate the great potential of the proposed ESOMPC in enhancing the temperature control accuracy of MPTL in terms of thermal management performance.

1. Introduction

As the electronic equipment becomes more sophisticated and precise, efficient temperature control requirement of high heat-flux electronic equipment promotes the booming development of the mechanically pumped two-phase loop (MPTL) technology [1-3]. Differing from the conventional two-phase heat dissipation techniques, such as heat pipe [4], thermosyphon [5] et al., MPTL provides larger heat transfer capacity due to the deployment of external driving source [6], which greatly upgrades the heat dissipation capacity of the cooling system [7]. Furthermore, the increasing density of the integration level requires a more efficient temperature management strategy for the electronic devices. In this context, the mini-channel evaporator, with the aggregated superiorities of nucleate boiling and forced convection, is of the prominent deliberation for replacing traditional evaporators and reinforcing heat transfer capacity [8,9]. By configuring externally driven source and mini-channel evaporator, MPTL could provide satisfactory temperature level and uniform temperature field of evaporator surface, where mini-channel is valuable for high heat-flux applications with small surface areas or the requirement of stable conditions (e.g., minimal temperature oscillation). To this end, as one promising scheme, the hybrid mini-channel evaporator and MPTL tends to be applied to the high-performance heat dissipation of precise electronic devices.

However, the stability of controlled temperature under uncertainty is of the prominent concern, leading to impede wider application of MPTL [10]. The operating temperature is considered to be a key indicator in terms of the application of MPTL, especially for fragile aerospace equipment [11]. Generally, wide range temperature variation, triggered by heat load fluctuation, especially deviating from the limit, will lead to undesirable thermal stress, and thus impairing the durability of

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https://doi.org/10.1016/j.applthermaleng.2022.118663
Received 22 August 2021; Received in revised form 29 April 2022; Accepted 14 May 2022
Available online 20 May 2022
1359-4311/

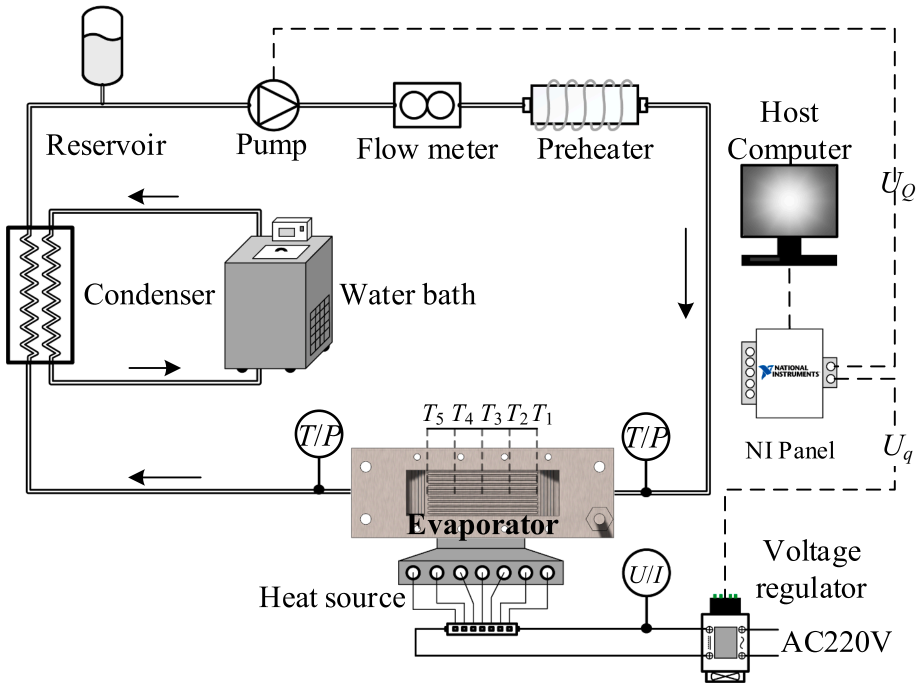


Fig. 1. A schematic view of the considered MPTL experimental platform.

instruments. Therefore, from the perspective of sustainability and efficiency, it is desirable to stabilize the controlled temperature under operation. A strong temperature control capability can enable the MPTL to be insensitive to variable load operation as to be applied to wider-range heat dissipation task with more flexibility and enhance the thermal stability of sensitive electronics equipment.

As the utilization of MPTL grows in the field of electronics, there have been several attempts devoted to efficient temperature regulation. An MPTL testbed, carried by cargo spacecraft (TZ1), is applied to test the on-orbit thermal performance, and the on-orbit experimens verify the gravity independence of MPTL and achieve temperature control accuracy of ± 0.5 °C in terms of temperature tracking [12], where an importance was emphasized to avoid temperature fluctuation induced during start-up/shut down process. Moreover, focusing on minimizing the temperature oscillation and mitigating the temperature overshoot of MPTL, an active flow control scheme is designed to smooth the temperature response during cold-start of low heat input [13]. Marcinichen et al. [14] and Zhang et al. [15] concentrated on the precision of temperature control of MPTL, where the temperature control strategies are proposed with considering several set values of controlled variables. Satisfactory results were obtained in both temperature homogeneity and stability, and thus leading to performance optimization in both reutilization of waste heat and system cooling capacity.

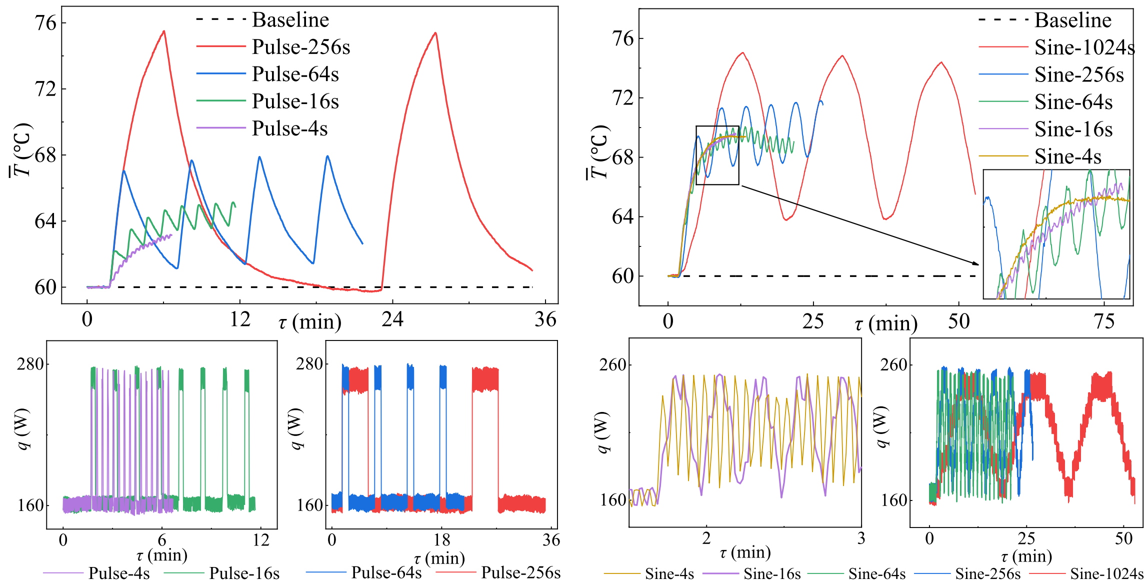
Nowadays investigation on MPTL control design mainly focuses on the temperature setpoints tracking [16]. However, temperature stabilization under heat load fluctuation is a relevant, yet challenging problem in practice, which usually occurs in the start-up process and at the variable-load operation and may deteriorate the operation of electronics equipment. Meanwhile, the improvement of MPTL temperature control accuracy is tricky primarily owing to following inevitable yet often unpredictable phenomena:

- Distinct temperature responses on flow rate and heat load, leading to a potentially inconvenient and relatively lagging effect of manipulated variable on disturbance compensation and thus potential degradation of the temperature control accuracy.
- Measurement noise, especially the output noise and the measured disturbance noise, which is an underlying and intractable difficulty for temperature controller.
- Various uncertainties caused by mini-channel boiling nonlinearity, mini-channel boiling is a coupling process of nucleate boiling and forced convection, with strong nonlinearity and uncertainty.

Such discrepancy between the potential performance increase of MPTL through mitigating the influence of heat load fluctuation on temperature and the current difficulties of temperature control on mini-channel boiling constitute the main motivation of this work. Therefore, this paper focuses on the temperature stabilization of MPTL, which is subjected to heat load disturbance fluctuation and mini-channel boiling nonlinearity. Considering that the model predictive control (MPC) possesses excellent control capabilities [17], and that the extended state observer (ESO) can accommodate the uncertainties well [18], both MPC and ESO have been successfully applied in industrial process control [19-21] and have great potential in optimizing the temperature stabilization control during heat load fluctuation [22,23]. The fluctuating heat load is regarded as the measurable disturbance, which can be used as augmented information to mitigate the temperature fluctuation via predictive model [24,25]. Furthermore, unmodeled dynamics and other unmeasurable disturbances/uncertainties are collectively considered as the “total disturbance” of the system and on-line reconstructed in the ESO to achieve non-offset control objectives.

By this means, an extended state observer-based model predictive controller (ESOMPC) is developed to treat heat load disturbance as a measurable signal to mitigate the temperature fluctuation and accompanied with the total disturbance to be reconstructed by the ESO while compensated via calculating manipulated variable. Compared with the current control solutions for MPTL, several benefits can be obtained by introducing the proposed method:

I. A relatively limited amount of calculation for the receding-horizon optimization, providing wider flexibility for real application of MPTL in terms of high-heat-flux cooling field.



(a) pulse wave disturbance (b) sine wave disturbance

Fig. 2. Temperature fluctuations under heat load disturbance.

II. The heat load fluctuations are used in the optimization algorithm as feed-forward information to be compensated in real-time, which is beneficial for temperature stabilization.
III. The uncertainties in terms of flow boiling nonlinearity and unequal response rates between process and disturbance, serve as an extended state variable incorporated into the extended state observer, leading to a disturbance rejection control objectives without any direct integration in the controller.

The aim of the proposed solution is to provide increased flexibility for MPTL to adapt to time-varying heat dissipation demand of electronic equipment in terms of heat load fluctuation, with the main contributions as follows:

1) An importance was emphasized to the issue of temperature variation under heat load fluctuation, which is devastating to the cooling system but neglected in state-of-the-art literature.
2) The nonlinearity of mini-channel boiling is analyzed based on the detailed open-loop experiments in order to provide better mitigation of the effect of mini-channel nonlinearity to temperature control performance.
3) The prominent operation issue in terms of heat load fluctuation is resolved by adopting ESOMPC as the controller for MPTL temperature stabilization control, depicting ESOMPC is a prospective method that can simultaneously achieve the goals of flow boiling nonlinearity resolving and heat load disturbance mitigation.
4) This paper show a promising prospective of proposed thermal control method in accelerating further application of the MPTL, leading to a dynamic temperature stabilization effect and thus prolonging the durability and lifetime of electronic equipment.

2. Problem formulation model identification

2.1. Experimental setup: Mechanically pumped two-phase loop

For research purposes, an experimental platform for mini-channel mechanically pumped two-phase loop (MPTL) is fabricated as shown in Fig. 1. Each channel of the 9-parallel rectangular mini-channel evaporator has a width of 1.5 mm, a height of 0.5 mm and a length of 45 mm. Besides the mini-channel evaporator combined with heat source as composite structure served as heat sink, other indispensable components are auxiliary for sustainable operation, such as a mechanical pump (brand: A500-4 N, uncertainty: ±2%), a gear flow meter (brand: TDS-100 h, uncertainty: ±1%), a reservoir, a preheater, data acquisition instrument (brand:NI-6002, uncertainty: ±0.6%), and a condenser cooled by an isothermal water bath (brand: XODC-2030-II, uncertainty: ±0.1 K). The voltage regulator is introduced to realize regulation of heat load imposed on evaporator. The working fluid is pumped by the mechanically pump while the flow rate is measured by flow meter. A liquid storage tank is working as the reservoir to store the redundant working. Five thermocouples are evenly arranged along the flowing direction on the evaporation surface below the mini-channel to measure the temperature of evaporation. Methanol, the latent heat of which is of 1109 kJ/kg at the saturation temperature of 64.7 °C (1 bar), is served as the working fluid.

Fig. 1 also concisely illustrates the indispensable equipment and analog signals related to the communication between each component and computer. The signal/data transmission between computer and actuators/sensors is based on the communication panel (National Instruments USB-6002). To this end, the speed of the mechanical pump and the heating voltage of heat source are adjusted through an analog voltage preset via computer. The current/voltage of heat load and the flow rate of the fluid are collected based on MODBUS-RTU protocol, which is omitted in the figure for clarity. The thermoelectric signals of thermocouples are measured via NI panel.

From the perspective of real-time control, the controlled variable is the average temperature ($\overline{T} = (T_1 + T_2 + T_3 + T_4 + T_5)/5$) of the evaporator and the manipulated variable is the analog voltage for pump adjustment (U_Q), while the analog voltage regulating the heat load is denoted as the disturbance variable (U_q). The heating power (q) and the flow rate (Q) are considered as intermediate variables. During the control experiments, the sampling frequency was set to 1 Hz to keep communication in a real-time regime in order to ensure the desired control quality.

2.2. Problem formulation

With the ever-increasing attentions to MPTL, optimal steady-state operational conditions, including optimal flow rate, have been unearthed. However, it is necessary to sacrifice the optimal parameter

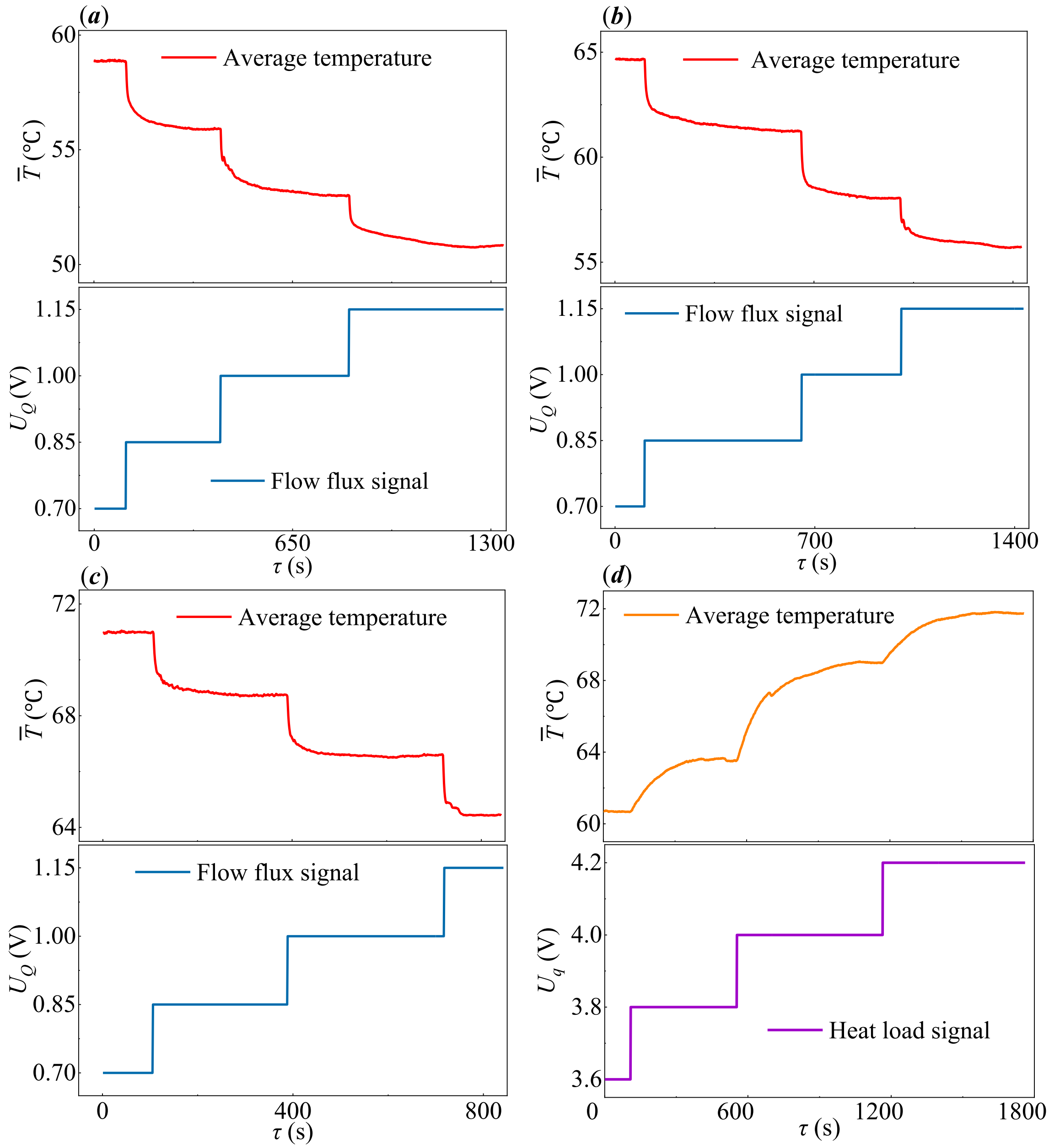


Fig. 3. Open-loop temperature response under flow rate step / heat load step, where (*a*) flow rate step at $q = 180$ W; (*b*) flow rate step at $q = 235$ W; (*c*) flow rate step at $q = 285$ W; (*d*) heat load step at $Q = 250$ mL/min.

settings to meet time-varying heat dissipation demand. In most cases, the heat load fluctuation with fast frequency and large amplitude could trigger corresponding devastating temperature oscillation. Therefore, it is desirable (for long-term operation) to regulate flow rate to stabilize the temperature under heat load fluctuation.

As mentioned above, it is mandatory to suppress the detrimental effects caused by the heat load fluctuation. However, regulation process of temperature stabilization is quite challenging, considering that the heat transfer of the boiling flow is the coupled result of nucleate boiling and forced convection, since both flow rate and heat load fluctuation will cause irregular and nonlinear impact.

2.2.1. *The effect of heat load fluctuation*

The operation of high heat-flux devices, such as radar, electronics communication base station, and high-energy weapon is often accompanied with variable load operation. From the perspective of heat dissipation, operation mode alteration implies heat load fluctuation and leads to temperature oscillation. To intuitively depict the above temperature deterioration, several background experiments on MPTL are implemented here. The flow rate is settled as constant while the fluctuations of heat load are exacerbated. The heat load variations are in the form of half sine wave and pulse wave to simulate the actual operation. Furthermore, the duty cycle of pulse wave heat load disturbance is settled as 20% to simulate start-up and stop process, and all the disturbances are under different response periods. The fluctuations of the average temperature of evaporation under differing periodic heat load disturbance are sketched in Fig. 2, where the time in the legend denotes the period of the heat load disturbance.

Following observations are made based on the figures. i) In the presence of heat load fluctuation without rapidly regulating the flow rate, the thermal performance of temperature management could severely deteriorate, which is characterized by the increment of evaporator average temperature (even by 16 °C in one case). ii) The response of temperature towards heat load disturbance is periodic and associated

Table 1
Identified plant model transfer functions at $q = 180$ W.

$G(s)$/Operation condition	U_Q (V)	k_p	T_z	T_w	ε
$k_p\frac{1}{1+2\varepsilon T_w s+(T_w s)^2}(1+T_z s)$	0.7–0.85	−19.97	41.56	18.58	1.93
	0.85–1	−19.19	40.63	17.28	2.55
$q = 180$ W	1–1.15	−15.86	108.99	28.72	3.45
$k_p\frac{1}{1+2\varepsilon T_w s+(T_w s)^2}(1+T_z s)$	0.7–0.85	−23.46	147.11	36.11	2.92
	0.85–1	−21.89	107.35	25.89	2.57
$q = 230$ W	1–1.15	−16.75	133.59	43.86	2.73
$k_p\frac{1}{1+2\varepsilon T_w s+(T_w s)^2}(1+T_z s)$	0.7–0.85	−15.01	53.96	18.44	1.97
	0.85–1	−14.62	32.51	13.57	1.75
$q = 285$ W	1–1.15	−19.53	107.59	24.91	3.11

Table 2
Identified plant model transfer functions at $Q = 250$ mL/min.

$f(s)$	U_q (V)	k_p	T_p	d
$k_p\frac{1}{1+T_p s}e^{-ds}$	3.6–3.8	15.29	106.70	−10.87
	3.8–4	27.56	131.23	−2.34
	4–4.2	14.66	139.55	−7.92

with the period of disturbance wave. It is predictable that the rapid fluctuation of temperature field will cause severe thermal stress. iii) It is difficult to reach the thermal steady-state under periodic heat load disturbance, which may lead to the wrongly identified model due to system nonlinearity and time-varying uncertainty, and thus create serious impediment for temperature control under the above thermal destabilization. iv) Moreover, the measurement noise is a nonnegligible obstacle that may impede the improvement of temperature control accuracy.

2.2.2. Nonlinearity of boiling flow

It is inferable that the nonlinear trends are encountered in mini-channel boiling, which is subject to fierce coupling between hybrid nucleate boiling and forced convection. Several open-loop flow rate step experiments and heat load step experiment are carried out to probe nonlinearities. The open-loop temperature step responses toward flow rate and heat load are shown in Fig. 3. A series of plant transfer function models in terms of various heat load conditions and a series of disturbance models in terms of differing flow rate conditions are obtained through data-driven identification. A nominal overdamped second-order transfer function model $G(s)$ and a typical first-order linear time-invariant (LTI) transfer function model $f(s)$ are introduced to identify the objective model and disturbance model, respectively. It should be emphasized that the identified linear models are valid only in certain operating conditions, and are not describing the generally nonlinear model of the object.

The mathematical models of $G(s)$ and $f(s)$ are respectively expressed as

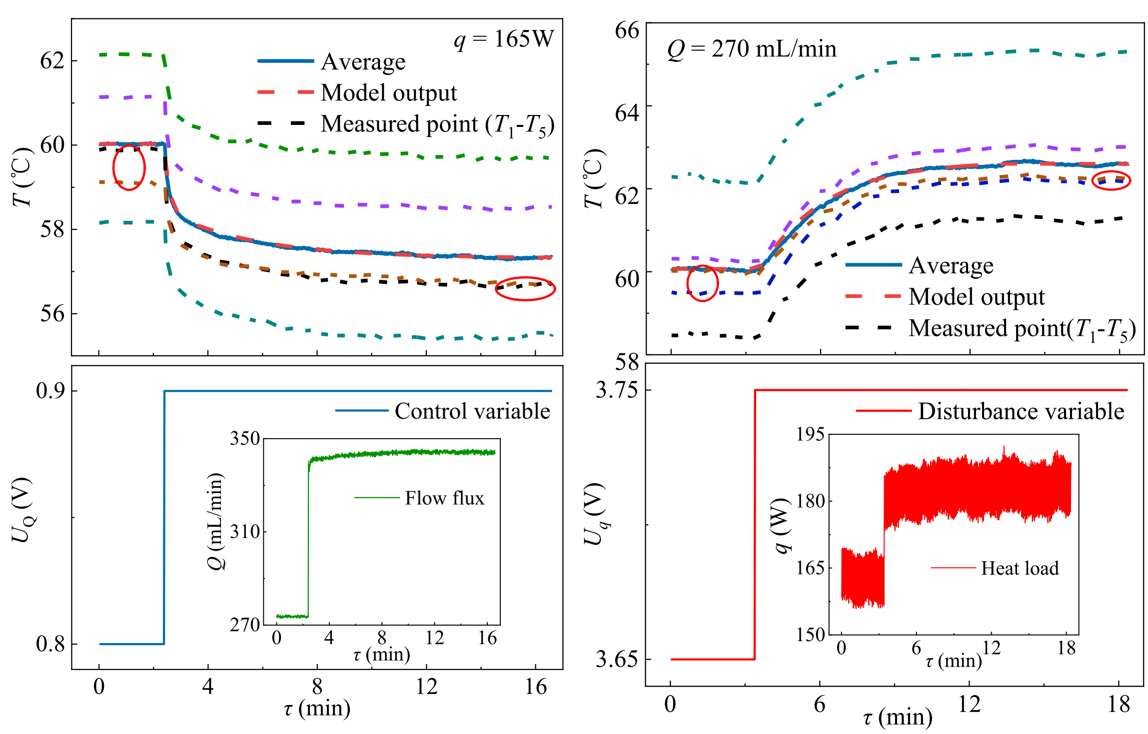
$$G(s) = k_p\frac{1}{1+2\varepsilon T_w s+(T_w s)^2}(1+T_z s) \tag{1}$$

$$f(s) = k_p\frac{1}{1+T_p s}e^{-ds} \tag{2}$$

where T_z and T_w are the time constant of the first-order differential link and the second-order oscillation link, respectively. ε denotes the damping factor, k_p is the proportional gain, T_p is the time constant of the first-order inertia link, and d denotes time-delay.

The identified plant model transfer functions under different cooling and heating operation situations are gathered in Tables 1 and Table 2, respectively. The identification results corroborate the fierce flow boiling nonlinearity. The diverse model parameters show disparate temperature responses toward flow rate or heat load, in both system inertia and time-delay, which indicate the irregular temperature responses under different operating conditions. This emphasizes the difficulty of flow boiling temperature control.

Besides the flow boiling nonlinearity, a pivotal control difficulty



(a) Process model G_Q(s)   (b) Disturbance model G_q(s)

Fig. 4. The fittness between the dynamic variation of temperature responses and the identified model $G_Q(s)$ and $G_q(s)$.

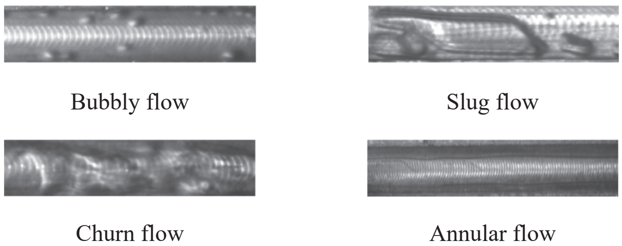


Fig. 5. The flow patterns observed in the minichannel.

comes from the responses delay between temperature response towards flow rate and towards heat load, which is quite challenging for heat load disturbance rejection. Other control difficulties consist of measurement noise and time-varying characteristic of mini-channel boiling. Therefore, the proposed advanced control algorithm could overcome the above obstacles aiming to realize satisfactory performance of heat load disturbance rejection.

2.3. Identification and nonlinearity analysis

For controller design and parameters tuning, data-driven identifications are conducted in this paper to identify the typical transfer function models of objective $G_Q(s)$ (i.e. average temperature response towards flow rate step) and disturbance $G_q(s)$ (i.e. average temperature response towards heat load step). The accuracy of data-driven identification is validated in comparison of the temperature step responses of experiments and the fitted transfer function models, as shown in Fig. 4, where solid line denotes average temperature, dashed line denotes temperature of measured point as shown in Fig. 1 and dotted line denotes temperature of identified model output.

(1) Process model:

$$G_Q(s) = -27.122\frac{1+120.99s}{(1+198s)(1+11.582s)} \tag{3}$$

(2) Disturbance model:

$$G_q(s) = \frac{25.978}{1+158.2s}e^{-10s} \tag{4}$$

It can be observed via the circle marked in the figure that the phenomenon of variation of temperature distribution along the flowing direction occurs during small range step, implying strong nonlinearity of mini-channel flow boiling. Moreover, through the visualization window on the microchannel evaporator, the transformation of flow patterns during minichannel boiling can be observed, which are shown in Fig. 5, which indicates the nonlinear heat transfer intensity.

3. Control design and tuning

As mentioned above, it is mandatory for the temperature stabilization of MPTL under heat load disturbance, which is conducive for electronics equipment, to be kept within optimal thermal state for safety and efficiency. However, the inevitable uncertainties resulted from the mini-channel boiling nonlinearity impede the temperature control. Furthermore, the unequal response rates between temperature towards flow rate and temperature towards heat load impose excess difficulty on temperature control under heat load fluctuation. Controller design and parameters tuning under such uncertain environment require advanced paradigm.

3.1. Conventional PI controller and feedforward compensator design

The choice of PI over PID is validated here by its satisfactory control performance in related literature and by the insensitivity of PI controller to measurement noise. The mathematic model of PI controller is:

$$U_Q(\tau) = h_p\left(e(\tau) + \frac{1}{T_i}\int_0^\tau e(\tau)d\tau\right) \tag{6}$$

where $e(\tau)$ is the error between the set-value and the system output, $U_Q(\tau)$ is the manipulated variable, h_p is the proportional gain, T_i is the integral time constant, and $k_i = h_p/T_i$ is the integral gain. Based on the SIMC-PI tuning algorithm [26], with the aim of minimizing the stabilization time, the parameters of PI controller are tuned as: $h_p = -0.0539$, $k_i = -0.012$.

A static feedforward compensator is embedded into the PI controller to tackle the abrupt change of heat load. The mathematic model of the static compensator is:

$$G_F(s) = -\left.\frac{G_Q(s)}{G_q(s)}\right|_{s=0} = 1.044 \tag{7}$$

3.2. Extended state observer-based model predictive controller

Aiming to tackle the flow boiling nonlinearity and heat load disturbance, this paper introduces an extended state observer-based model predictive controller (ESOMPC) to achieve non-offset temperature control objectives. It is done via structuring disturbance embedded predictive model and lumping the unmodeled disturbances as the total disturbances to the ESO, which can effectively solve the problem of both mini-channel nonlinearity and heat load fluctuation disturbances.

In this work, a following class of linear, time-invariant discrete SISO systems is considered

$$\begin{cases} x_{k+1} = Ax_k + BU_k \\ y_k = Cx_k \end{cases} \tag{8}$$

where x_k and y_k denote the system state and output, respectively, U_k is the system manipulated variable, A, B and C are the system matrices, and the subscript k denotes the time instant.

Remark 1. It is straightforward to show that in a specific case, the identified transfer functions of the MPTL system from Sect. 2.2 can be represented as corresponding models in form of (8).

Aiming to achieve the performance of external measurable disturbance compensation, a system model with the disturbance term is considered as follows:

$$\begin{cases} \begin{bmatrix} x_{k+1} \\ d_{k+1} \end{bmatrix} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix}\begin{bmatrix} x_k \\ d_k \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix}U_k + w_k \\ y_k = [C \ C_d]\begin{bmatrix} x_k \\ d_k \end{bmatrix} + v_k \end{cases} \tag{9}$$

where d_k is the measurable disturbance, B_d and C_d are the matrices denoting the impact of disturbance imposed on the system state and output, and w_k and v_k are noise disturbances for the state equation and the output equation, respectively.

An extended state model is designed as follows

$$\begin{cases} \begin{bmatrix} x_k \\ d_k \\ f_k \end{bmatrix} = \begin{bmatrix} A & B_d & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}\begin{bmatrix} x_{k-1} \\ d_{k-1} \\ f_{k-1} \end{bmatrix} + \begin{bmatrix} B \\ 0 \\ 0 \end{bmatrix}U_k + w_k \\ y_k = [C \ C_d \ 0]\begin{bmatrix} x_k \\ d_k \\ f_k \end{bmatrix} + v_k \end{cases} \tag{10}$$

where the augmented term f_k is as treated as the “total disturbance”.

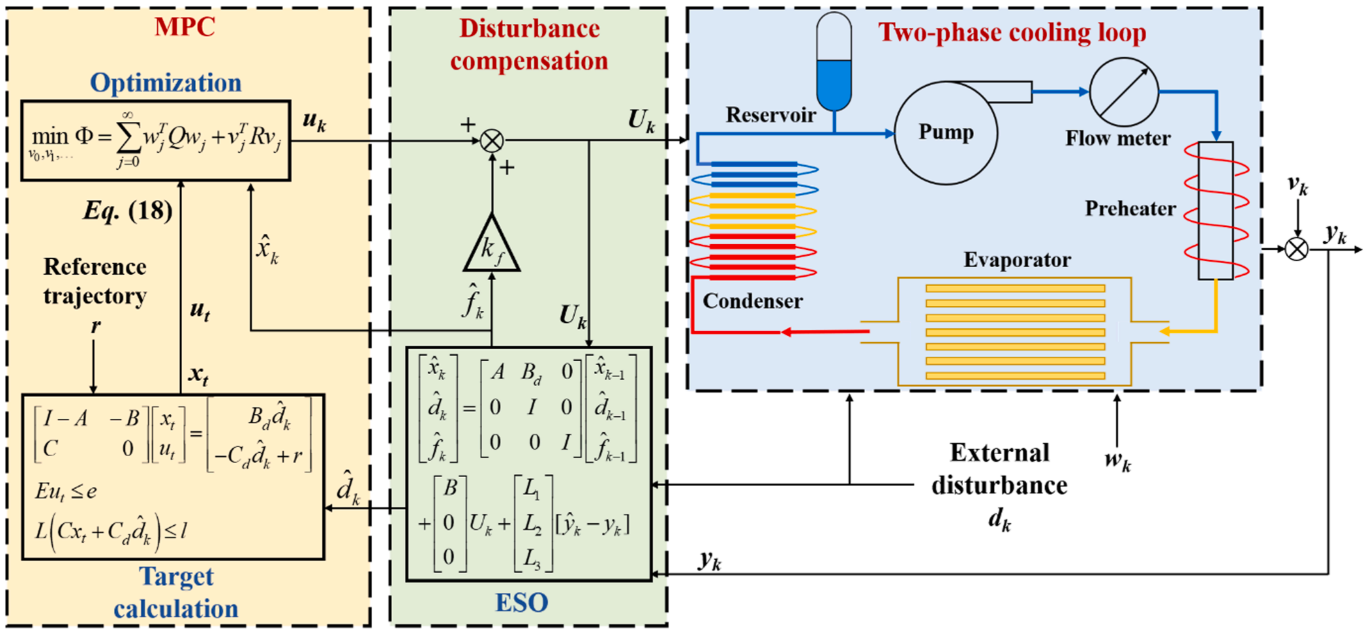


Fig. 6. Control structure of the proposed ESOMPC for MPTL temperature control.

Based on the extended state model (10), a following linear ESO can be designed for estimating the state variables (including the total disturbance) and it serves as a filter for the measured output and disturbance:

$$\begin{cases} \begin{bmatrix}\hat{x}_k\\ \hat{d}_k\\ \hat{f}_k\end{bmatrix} = \begin{bmatrix} A & B_d & 0\\ 0 & I & 0\\ 0 & 0 & I\end{bmatrix}\begin{bmatrix}\hat{x}_{k-1}\\ \hat{d}_{k-1}\\ \hat{f}_{k-1}\end{bmatrix} + \begin{bmatrix}B\\ 0\\ 0\end{bmatrix}U_k + \begin{bmatrix}L_1\\ L_2\\ L_3\end{bmatrix}[\hat{y}_k - y_k] \\ \hat{y}_k = [C\ C_d\ 0]\begin{bmatrix}\hat{x}_k\\ \hat{d}_k\\ \hat{f}_k\end{bmatrix}\end{cases} \tag{11}$$

where L_1, L_2 and L_3 are the observer gains for the estimated state, estimated external disturbance, and estimated total disturbance, respectively. For simplicity of tuning, the observer gain vector is determined by:

$$L_i = M^{-1}N_i \tag{12}$$

where M and N_i can be obtained by solving the following linear matrix inequality [27]:

$$\begin{bmatrix} M^T + M - W & * \\ MA_i - N_iC_i & W\end{bmatrix} > 0,\ W = W^T > 0 \tag{13}$$

Remark 2. It is not difficult to show that in a specific case, the nonlinear, perturbed MPTL temperature control system can be represented in the form of (11) when lumping all the nonlinearities and acting perturbations into a single disturbance term and making it an extra, virtual system state variable. Such aggregation into a sole "total disturbance" term, allowing its straightforward estimation and compensation, is a trademark of control methods based on the idea of active disturbance rejection control (ADRC), which now have a track record of successful applications in power, process, and motion control areas [18,28,29]. In our previous study in [30], the efficacy of ESO in disturbance estimation has been verified in a theoretic manner.

The estimation of system state $\hat{x}_k$ and the estimated external disturbance $\hat{d}_k$ are served as the input parameters for MPC optimization, the calculation of which is based on the following quadratic program [31]:

$$\min_{x_t,u_t} (u_t - \overline{u})^T \widetilde{R}(u_t - \overline{u}) \tag{14}$$

subject to

$$\begin{cases} \begin{bmatrix} I - A & -B \\ C & 0\end{bmatrix}\begin{bmatrix}x_t\\ u_t\end{bmatrix} = \begin{bmatrix} B_d\hat{d}_k \\ -C_d\hat{d}_k + r\end{bmatrix} \\ E_iu_t \leqslant e_i \\ L(Cx_t + C_d\hat{d}_k) \leqslant l \end{cases} \tag{15}$$

where r and $\overline{u}$ are the set value for the controlled and manipulated variables, respectively, E_i and e_i are usvsed to denote input constraints, and L and l describe output constraints, respectively.

Based on the above, the manipulated variable can be calculated by a following infinite-horizon optimization function [32]:

$$\min_{u_0,u_1,\cdots} \Phi = \sum_{k=0}^{\infty} (y_k - r)^T Q(y_k - r) + (u_k - u_t)^T R(u_k - u_t) \tag{16}$$

subject to input and output constrains:

$$\begin{cases} E_iu_k \leqslant e_i\ k = 0,\ 1,\dots \\ Ly_k \leqslant l\ k = 0,\ 1,\dots \end{cases} \tag{17}$$

where Q and R are the weighting matrices on controlled and manipulated variable, respectively. The deviation variables can be calculated based on (10) as

$$\begin{cases} w_j = \hat{x}_{k+j} - x_t \\ v_j = u_{k+j} - u_t \end{cases} \tag{18}$$

the original optimization objective (16) becomes

$$\min_{v_0,v_1,\cdots} \Phi = \sum_{j=0}^{\infty} w_j^T Q w_j + v_j^T R v_j \tag{19}$$

subject to

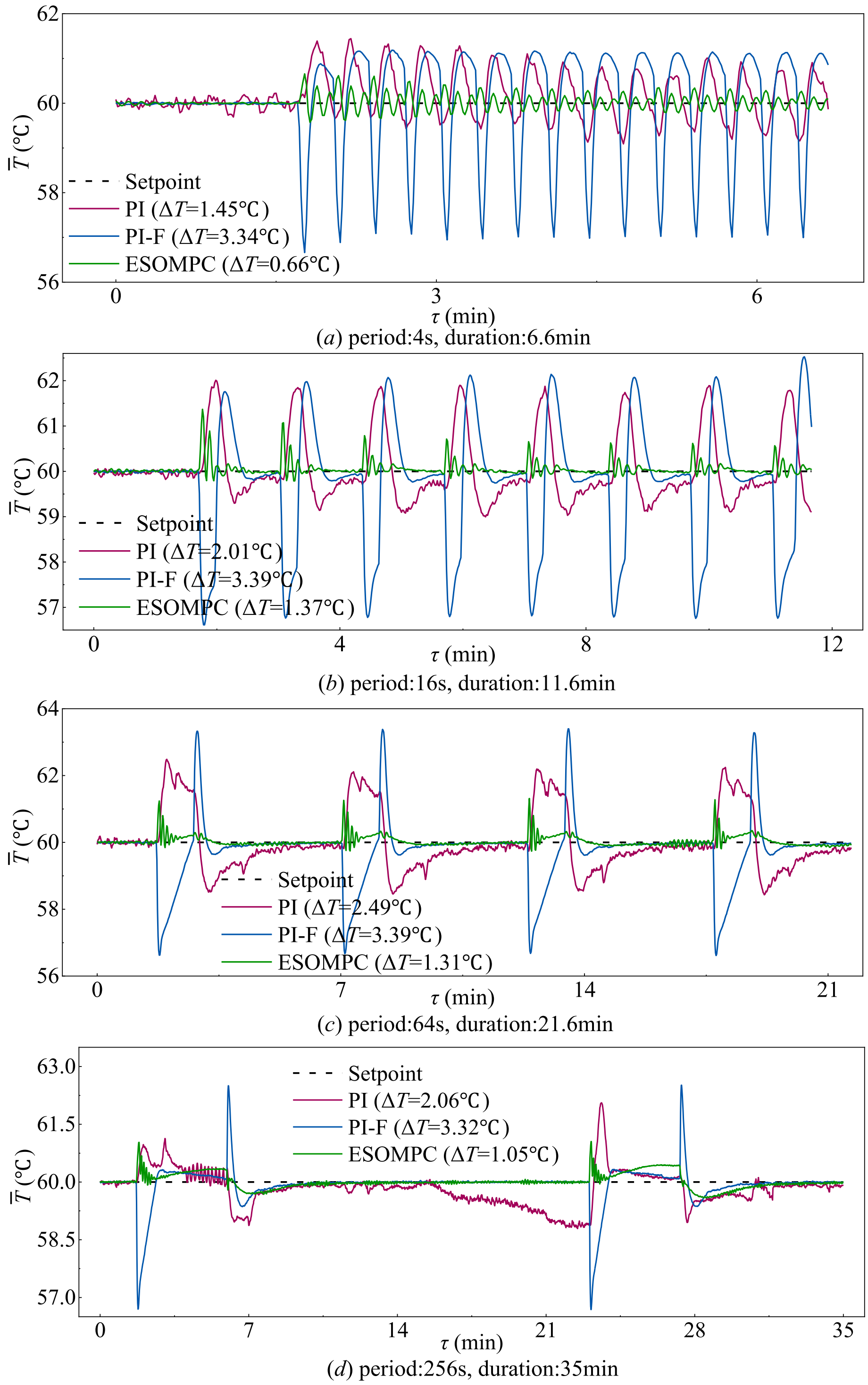


Fig. 7. Dynamic wall temperatures under pulse-wave heat load by different self-control approaches.

$$\begin{cases} w_{j+1} = Aw_j + Bv_j \; k = 0,\ 1, \ldots \\ E_i u_j \leqslant e_i - Eu_t \; k = 0,\ 1, \ldots \\ LCw_j \leqslant l - L(Cx_t + C_d \widehat{d}_k) \; k = 0,\ 1, \ldots \end{cases} \tag{20}$$

Furthermore, the estimated total disturbance $\widehat{f}_k$ can then incorporated to the manipulated variable of MPC as a feedforward signal and provide better mitigation of the influence of nonlinear disturbances and/or modeling mismatches. By this means, the proposed ESOMPC can reach a reasonable trade-off between set-points tracking and disturbance rejection.

$$U_k = u_k + k_f \widehat{f}_k \tag{21}$$

where u_k is the designed ESOMPC output, k_f is the total disturbance compensation gain[27], designed as

$$k_f = -(\Theta B + D)^{-1} \tag{22}$$

where $\Theta = (C + D)(I - A - B)^{-1}$.

Remark 3. The convergence of (14) and (15) was proven in [33] and

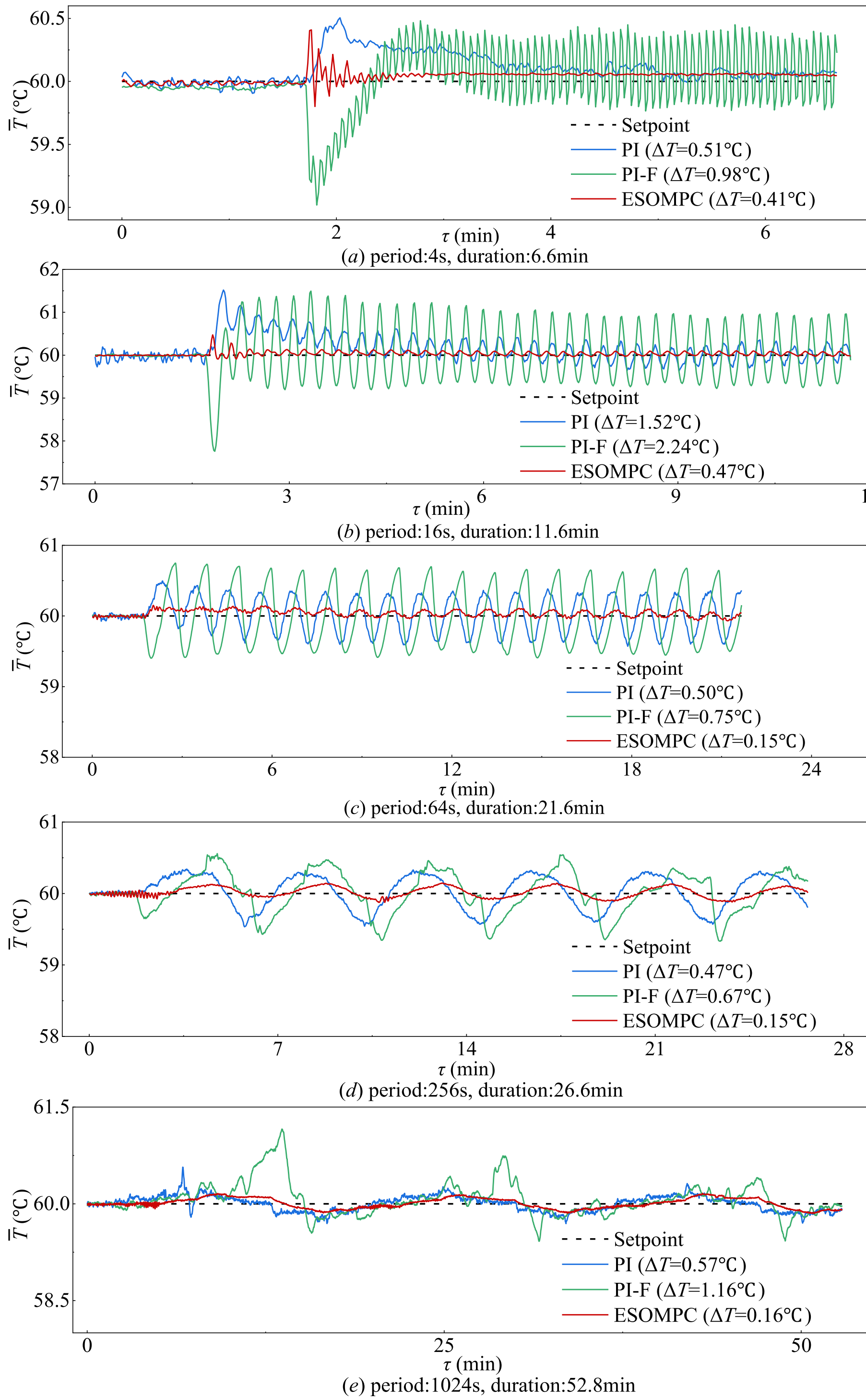


Fig. 8. Dynamic wall temperatures under sine-wave heat load by different self-control approaches.

the stability of ESOMPC can be extended to prove local Lyapunov stability of the closed-loop system (11) under standard regularity assumptions on the state update function f in [34].

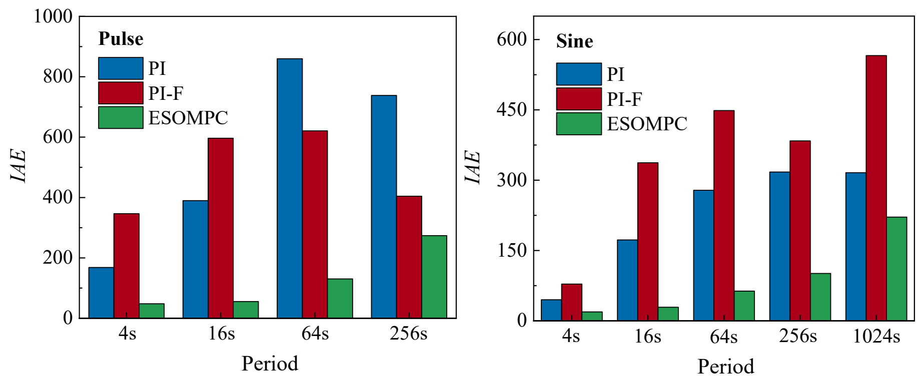
The optimization procedure of objective function (Eq. (19)) is discussed in [35] and [36], where the infinite-horizon constrained problem is reparameterized in terms of a finite number of decision variables and constraints by appending the optimal LQR unconstrained control law.

So far, the deductions of ESOMPC are accomplished with the control structure shown in Fig. 6. With the aim of minimizing the stabilization time towards the quasi-steady state, the parameters tuning is done manually resulting in the sampling time being set as $T_s = 1$ s, prediction horizon as $P = 110$, the control horizon as $M = 6$, the error weighting matrix of the temperature as $Q = 0.18$, and the control weighting matrix as $R = 0.45$. For safe operation of mechanical pump, the manipulated variable is constrained at $U \in [0, 1.3]$.

4. Experimental results and discussion

4.1. Experimental results

The comparisons between the PI controller, the hybrid PI controller



(a) Pulse-wave heat load (b) Sine-wave heat load

Fig. 9. Comparison of IAE results.

with feedforward compensator (PI-F), and the proposed ESOMPC approach in terms of control performance are conducted under identical steady-state setting with equivalent disturbance variation (i.e. amplitude, period, total duration), as shown in Section 2.2.1. The experiments are carried out by making the system reach its steady state first, then the heat load disturbances (in form of half-sine wave or pulse wave) are imposed on the mini-channel evaporator. The quality of disturbance rejection of PI, PI-F, and the proposed ESOMPC is compared in Fig. 7 (for pulse wave disturbance) and Fig. 8 (for sine wave disturbance), where the label 'PI' denotes the conventional PI controller, 'PI-F' denotes the hybrid PI controller with feedforward compensator, and 'ESOMPC' denotes the proposed approach. The test duration and the period of disturbance are marked in the figures. As shown in results, ESOMPC shows superior temperature stabilization performance than PI and PI-F controller, both in temperature fluctuation magnitude and in settling time.

4.2. Discussion

For intuitive judgement of the control performances of temperature stabilization, an integral quality index IAE (integral absolute error) is applied and defined as

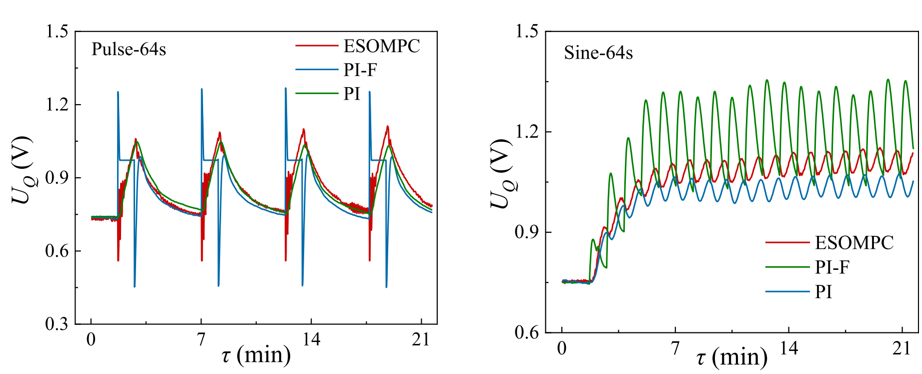
$$IAE = \int_0^\tau |y(\tau) - r| d\tau \tag{23}$$

where τ is time, $y(\tau)$ is the real-time average temperature of the evaporation surface, and r is the set value of average temperature. The synthetic uncertainty of IAE is same as the uncertainty of measuring temperature T, which is attributed to the linear relation between IAE and T. Thus, the synthetic uncertainty of IAE is $\pm$ 0.5 °C. Moreover, ΔT is used to denote the temperature fluctuation amplitude. ΔT is defined as $\Delta T = |T_m - r|$, where T_m is the extremum temperature during control experiments.

Generally, the smaller both the IAE value and ΔT, the smaller the deviation of average temperature from the set value. ΔT is marked in the legend in Fig. 7 and Fig. 8. Fig. 9 contain detailed comparison of PI, PI-F, and the proposed ESOMPC based on the corresponding IAE calculations.

From the perspective of quantitative evaluation, both IAE and ΔT under temperature control by ESOMPC are lower than those of other two controllers. Indicating that benefited from the real-time estimation of ESO, the ESOMPC is superior in terms of heat load disturbance rejection to the hybrid PI controller with feedforward compensator, which is verified through the smaller temperature deviation from the set value. By contrast, the nonlinearity characteristics and modelling mismatch of the flow boiling worsen the performance of the hybrid PI controller with feedforward compensator in terms of disturbance rejection. Larger temperature fluctuation magnitude and prolonged settling time indicate that the traditional PI controller and compensator based on the identified model are defective in such coupling process subjected to uncertainties. Even in some cases, PI is superior to PI-F, which can be attributed to the modelling mismatch resulted from mini-channel boiling. Furthermore, the experimental results verify the efficacy of ESO, which provides precise system state via real-time modification on predictive model.

Besides nonlinearity and modelling mismatch, the unequal response rates between process model (3) and disturbance model (4) impose additional difficulty on temperature control. The lead control of feedforward compensator and lag response of PI controller are inadequate for this, while the ESOMPC with embedded disturbance model is of stronger capability in dealing with such obstacle. It can be verified in Fig. 10 that the response rate of ESOMPC is between the PI and the



cf. Fig. 8(c) cf. Fig. 9(c)

Fig. 10. The control trajectories of manipulated variables.

hybrid PI with feedforward compensator, which leads to the superiority of ESOMPC.

The realization of non-offset goal of ESOMPC, shown in the comparison results, can be attributed to the estimation of unmodeled uncertainties and the effective disturbance compensation via designing the ESO and embedding the disturbance model into the predictive model, respectively.

5. Conclusions

In this paper, the extended state observer-based model predictive control (ESOMPC) design was proposed for temperature control of MPTL. Through a set of experiments, it was found that the introduced solution has the following advantages over standard PI and PI-F controllers: i) the criterion IAE of tracking errors between the evaporator temperature and set value can be reduced drastically; ii) the property of mitigating temperature oscillation caused by the heat load variation can be improved; and iii) the effective robustness against unmodeled uncertainty can be gained. Obtained smaller temperature fluctuation magnitude and shorter settling time indicate that the proposed ESOMPC has stronger capabilities in dealing with periodic heat load disturbance, measurement noise, and various unmodeled uncertainties. The experimental results in this paper show a promising prospective of ESOMPC in accelerating further application of the MPTL.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by the National Natural Science Foundation of China (NSFC) under Grant 51936003 & 51806034, Natural Science Foundation of Jiangsu Province, China under Grant BK20211563, and the open funding of Key Lab for Thermal Science and Power Engineering of Ministry of Education, Tsinghua University.

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